

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY**

INVITATIONAL B MEET • 2016

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|-------|-------|-------|
| 1. B | 21. D | 41. D |
| 2. C | 22. B | 42. C |
| 3. A | 23. A | 43. B |
| 4. E | 24. C | 44. A |
| 5. B | 25. B | 45. A |
| 6. B | 26. E | 46. D |
| 7. B | 27. A | 47. B |
| 8. C | 28. B | 48. D |
| 9. E | 29. D | 49. B |
| 10. B | 30. C | 50. A |
| 11. D | 31. E | 51. D |
| 12. E | 32. D | 52. B |
| 13. C | 33. A | 53. E |
| 14. D | 34. C | 54. C |
| 15. C | 35. E | 55. A |
| 16. E | 36. D | 56. C |
| 17. A | 37. A | 57. D |
| 18. B | 38. B | 58. E |
| 19. C | 39. E | 59. B |
| 20. B | 40. C | 60. A |

PHYSICS KEY for Science Contest • Invitational B Meet • 2016

41. (D) page 2: "The principle of science, the definition, almost, is the following: The test of all knowledge is experiment."
42. (C) page 20: "...there is nothing that living things do that cannot be understood from the point of view that they are made of atoms acting according to the laws of physics... it is the most useful theory for producing new ideas in the field of Biology."
43. (B) page 36: "... no phenomenon directly involving a frequency has yet been detected above approximately 10^{12} cycles per second."
44. (A) page 42: "... particles which do not interact strongly in nuclei, have nothing to do with the nuclear interaction... These are called leptons..."
45. (A) Blue and white stars are significantly larger, hotter and less common than red and orange stars. Because of their size and temperature they are much more luminous, but have very short life spans.
46. (D) Use $v = v_0 + at$ for the balloon, giving the upward velocity after 12.0 seconds as
 $v = 0 + 4.20 * 12 = 50.4 \text{ m/s}$. This is the initial velocity of the sandbag.
 Now use $x = x_0 + v_0t + \frac{1}{2}at^2$ to get the height of the balloon when the sandbag is dropped.
 $x = 0 + 0 * 12 + \frac{1}{2} * 4.2 * 12^2 = 302.4 \text{ m}$ This is the initial height of the sandbag.
 Again, use $x = x_0 + v_0t + \frac{1}{2}at^2$, this time for the sandbag.
 $x = 302.4 + 50.4t + \frac{1}{2}(-9.8)t^2 = 0$ (the final position of the sandbag is back at the ground).
 This gives the quadratic: $t^2 - 10.3t - 61.7 = 0$, solutions are $t = 14.5, -4.25$ seconds. Only the positive time solution is valid.
47. (B) Use the constant angular acceleration equation $\omega^2 = \omega_0^2 + 2\alpha\Delta\theta = 120^2 = 0 + 2(3.5)\Delta\theta$. This gives an angle of rotation of $\Delta\theta = \frac{14400}{7} = 2057 \text{ rad}$. Convert to revolutions:
 $\Delta\theta = \frac{2057}{2\pi} = 327 \text{ revolutions}$
48. (D) The resistance of the parallel group is $R_1 = (1/60 + 1/40)^{-1} = 24 \text{ ohms}$. So the total resistance of the circuit is $R_T = 50 + 24 = 74 \text{ ohms}$. The total current is given by $I_0 = 9.00/74 = 0.1216 \text{ A}$.
 This gives a voltage across the parallel group of $V_1 = I_0R_1 = (.1216)(24) = 2.92 \text{ V}$. This is the same voltage across each resistor in the parallel group. So the current through the 40 ohm resistor is given by: $I_{40} = V_1/40 = 2.92/40 = 0.073 \text{ A} = 73 \text{ mA}$.
49. (B) Using Newton's second Law: $F = ma$ for each box. Forces on the hanging box are only vertical:
 $F = T - mg = T - (6)(9.8) = T - 58.8 = -ma = -6a$. Acceleration is negative since it is downward. Vertical forces on the sliding box are: $F = N - mg = N - (2.5)(9.8) = N - 24.5 = 0$. Vertical acceleration is zero since the sliding box moves only horizontally. This gives the Normal force $N = 24.5$ Newtons. Horizontal forces on the sliding box are:
 $T - f = T - \mu N = T - (0.35)(24.5) = T - 8.575 = ma = 2.5a$. Solving for T in this equation:
 $T = 2.5a + 8.575$. Plug this into the equation from the hanging box:
 $T - 58.8 = 2.5a + 8.575 - 58.8 = -6a$.
 $-8.5a = -50.225$ or $a = 50.225/8.5 = 5.91 \text{ m/s}^2$. Plug this back into the tension equation:
 $T = 2.5a + 8.575 = (2.5)(5.91) + 8.575 = 23.2 \text{ Newtons}$.
50. (A) Use conservation of energy. The initial energy stored in the spring is
 $E = \frac{1}{2}kx^2 = \frac{1}{2}(4500)(0.15)^2 = 50.625 \text{ J}$
 This converts into gravitational potential energy and Work done by the drag force. The distance over which work is done equals the height to which the sphere ascends.
 $E = 50.625 = mgh + W = (0.6)(9.8)h + fh = 5.88h + 1.8h = 7.68h$
 Solving for height gives: $h = \frac{50.625}{7.68} = 6.59 \text{ m}$.

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51. (D) This scenario is the time reversal of a person jumping into a swimming pool; so all time-independent laws, such as conservation of energy and momentum, will still be obeyed. One law that is time-dependent is the Second Law of Thermodynamics – the law that states that entropy will not decrease, and generally increases, as time passes. Our time reversed scenario could only happen if entropy decreased, which is why we do not see such occurrences in everyday life.
52. (B) $f = 1/2\pi\sqrt{1/LC} = (1/2\pi)\sqrt{1/(400 \times 10^{-12})(0.1)} = 25200 \text{ Hz} = 25.2 \text{ kHz}$.
53. (E) The electric potential for one charge is given by $V = kQ/r$. It is not a vector, so direction doesn't matter, just the absolute distance between charge and point. Assume that the point on the y-axis is a distance y up from the origin. Then the distance to charge Q is y, and the distance to charge -2Q is $\sqrt{r^2 + y^2}$. Then $V = 0 = kQ/y + k(-2Q)/\sqrt{r^2 + y^2}$. So $kQ/y = 2kQ/\sqrt{r^2 + y^2}$, which leads to $2y = \sqrt{r^2 + y^2} \rightarrow 4y^2 = r^2 + y^2 \rightarrow 3y^2 = r^2 \rightarrow y = r/\sqrt{3}$
54. (C) The current in the railgun is $I = V/R = 800/2.5 = 320 \text{ A}$. Then the force on the projectile is $F = I\ell B = (320)(0.2)(0.3) = 19.2 \text{ N} = ma$, so the acceleration is $a = F/m = 19.2/(.005) = 3840 \text{ m/s}^2$. The velocity at the end of the rails is then $v^2 = 0^2 + 2(3840)(1.2) \rightarrow v = 96.0 \text{ m/s}$
55. (A) Refracting surface, object (fish) is in water, image is in air: $n_1 = 1.33$ and $n_2 = 1.00$. A bubble is convex as seen from the object (fish), so the sign of the radius is positive.
 $n_1/s + n_2/s' = (n_2 - n_1)/R =$
 $1.33/0.5 + 1.00/s' = (1.00 - 1.33)/1.2 \rightarrow 2.66 + 1/s' = -0.275 \rightarrow \frac{1}{s'} = -2.935$
 So $s' = -0.341 = -34.1 \text{ cm}$ or 34.1 cm from the bubble's edge.
56. (C) For diffraction $a \sin \theta = m\lambda$ where $\tan \theta = x/L$. The width of the central maximum equals $2x$, so $x = 3.66/2 = 1.83 \text{ cm}$. Then $\tan \theta = 1.83/200 = 0.00915 \rightarrow \theta = 0.5242 \text{ degrees}$. This is the angle for the first dark fringe, so $m = 1$. So, $a * \sin(0.5242) = (1)(577 \times 10^{-9})$
 $a = (577 \times 10^{-9})/(0.00915) = 6.31 \times 10^{-5} \text{ m} = 63.1 \mu\text{m}$
57. (D) $N = N_0 e^{-\lambda t} \rightarrow \lambda t = \ln(N_0/N) = \ln(22.5/18.3) = 0.2066 = \lambda(36)$.
 Then $\lambda = 0.00574 \text{ hours}^{-1}$. The half-life is then $t_{1/2} = \ln(2)/\lambda = 0.693/0.00574 = 121 \text{ hrs}$.
58. (E) Three quarks $\rightarrow$ Baryon. Charge is given by $u(+2/3) + u(+2/3) + s(-1/3) = +1$. Strangeness for one s-quark is $S = -1$.
59. (B) Conservation of momentum: $m_{\text{hammer}} v_{\text{hammer}} = m_{\text{astro}} v_{\text{astro}} = (1.5)(12) = 18 = (140) v_{\text{astro}}$
 Then $v_{\text{astro}} = 18/140 = 0.1286 \text{ m/s}$
 To get within reach of the spacecraft: $t = x/v = (4.2 - 0.5)/0.1286 = 28.8 \text{ seconds}$
60. (A) $KE = hf - \phi = hc/\lambda - \phi = (1240 \text{ eVnm})/(404 \text{ nm}) - \phi = 3.07 - \phi = 0.76$
 So $\phi = 3.07 - 0.76 = 2.31 \text{ eV}$